

RELATIONAL DATABASE VS. NoSQL DATABASE

ASSIGNMENT 4

DATABASE MANAGEMENT SYSTEM

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INTRODUCTION

Database technology is essential for managing information generated by online shopping applications. Relational and NoSQL databases provide different ways to store and retrieve data. This assignment compares their structure, scalability, consistency, performance, schema, relationships, integrity, and use cases with reference to the Online Shopping Management System.

COMPARISON BETWEEN RELATIONAL DATABASES AND NOSQL

DATABASES

A relational database stores data in tables consisting of rows and columns and uses SQL to create, retrieve, update, and delete records. Examples include MySQL, PostgreSQL, and Oracle Database.

A NoSQL database stores information using models such as documents, key-value pairs, columns, or graphs. Examples include MongoDB, Redis, and Apache Cassandra.

Feature	**Relational Database**	**NoSQL Database**
Data Structure	Tables with rows and columns.	Documents, key-value pairs, columns, or graphs.
Scalability	Often scales vertically; horizontal scaling is also possible.	Many systems are designed for horizontal scaling across servers.
Consistency	Typically supports strong consistency and ACID transactions.	Consistency depends on the product and configuration.
Performance	Effective for structured data, joins, and complex queries.	Effective for workloads suited to its data model and distribution.
Schema	Usually has a predefined schema.	Often supports flexible schemas.
Relationships	Primary keys, foreign keys, and joins connect tables.	Relationships may use embedded documents, references, or model-specific structures.
Data Integrity	Supports PRIMARY KEY, FOREIGN KEY, UNIQUE, and CHECK constraints.	Integrity mechanisms depend on the database and application.

Suitable Use Cases	Order processing, payments, inventory, and structured records.	Catalog content, caching, flexible product attributes, and distributed workloads.
Examples	MySQL, PostgreSQL, Oracle Database.	MongoDB, Redis, Apache Cassandra.

WHY ORGANIZATIONS CHOOSE DIFFERENT DATABASE TECHNOLOGIES

RELATIONAL DATABASES

Organizations prefer relational databases when information is structured and relationships must remain accurate. Online shopping orders connect customers, order items, products, payments, and deliveries. Relational databases are useful because:

1. Structured data can be organized into tables with defined columns and data types.
2. Primary and foreign keys help maintain referential integrity.
3. Transactions help ensure reliable order and payment processing.
4. SQL supports filtering, grouping, aggregate functions, and joins.
5. Inventory and order totals can be queried consistently.

NOSQL DATABASES

Organizations may choose NoSQL when they need flexible product attributes, horizontal scaling, or distributed data storage. Reasons include:

6. Flexible data models can represent product types with different attributes.
7. Horizontal scaling can distribute workloads across servers.
8. Document databases can store product descriptions and metadata in a flexible form.
9. Key-value databases can support caching and fast access patterns.

SUITABLE DATABASE CHOICE FOR THE ONLINE SHOPPING MANAGEMENT SYSTEM

A relational database such as MySQL is suitable for the core system because customer orders, payments, stock, and delivery records have clear relationships and require consistency. A NoSQL database may complement it for flexible catalog descriptions or caching, but it is not necessary for the basic assignment design.

CONCLUSION

Relational and NoSQL databases each have strengths. For the proposed Online Shopping Management System, MySQL provides a clear structure for related shopping records and integrity constraints. NoSQL can be useful for particular flexible or distributed workloads depending on future requirements.